

Observations for the Study – CareerQuestXR

1. Participant Engagement with CareerQuestXR

We will observe how participants interact with the CareerQuestXR system, focusing on their level of engagement during virtual reality aptitude and career exploration sessions. This includes:

- Responsiveness to immersive tasks (problem-solving, decision-making, planning tasks).
 - Comfort with wearing and using the VR headset.
 - Willingness to immerse themselves in different scenarios and role simulations.
 - Degree of persistence and focus across multiple VR tasks.
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2. User Experience and Ease of Use

We will monitor participant feedback and behaviour regarding the usability and accessibility of the system, including:

- How easily they navigate the VR environment and menus.
 - Their ability to complete tasks without confusion.
 - Any technical difficulties encountered (motion sickness, calibration issues, hand tracking problems).
 - User strategies for interacting with VR elements (e.g., trial-and-error vs. planned approaches).
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3. Career Path Awareness and Discovery

We will observe how participants' awareness of different career paths changes before and after using CareerQuestXR:

- Identification of new career options they had not previously considered.
 - Shifts in perceived “fit” for certain career domains after receiving system recommendations.
 - Confidence levels in making academic or career decisions post-interaction.
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4. Decision-Making Process and Self-Reflection

We will focus on how participants' decision-making processes evolve while interacting with CareerQuestXR:

- Whether their choices align more closely with their measured strengths, interests, and Big Five / RIASEC results.
 - Notable moments of self-reflection or re-evaluation of their current academic or career paths.
 - Strategies used to handle errors, frustration, or time pressure in tasks.
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5. Perceived Accuracy of Career Aptitude Matching

We will note participants' perceptions of the accuracy and relevance of CareerQuestXR's career recommendations, including:

- Whether they feel their interests, aptitudes, and personality traits are reflected in the results.
 - Reactions to the explanation text accompanying career recommendations (Big Five + RIASEC code).
 - Comparisons participants make between VR-based feedback and the SAT-style Google Form test.
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6. Attitudes Toward Virtual Reality for Career Guidance

We will explore participants' attitudes toward VR technology in the context of career guidance, noting:

- Comfort level with immersive experiences.
 - Openness to technology-driven aptitude testing compared to traditional formats.
 - Concerns about trust, fairness, or bias in the system's recommendations.
 - Preferences for VR vs. traditional assessments.
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7. Emotional Reactions and Behavioural Indicators

We will record emotional and behavioural responses during and after VR sessions:

- Signs of excitement, curiosity, stress, or frustration.
 - Coping strategies during challenging tasks.
 - Changes in self-reported stress before, during, and after tasks.
 - Observable differences between participants' reactions to Google Form tasks vs. VR tasks.
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8. Biometric and Self-Reported Data

Since biometric self-reporting (heart rate, stress levels) is compulsory, we will observe:

- How physiological indicators fluctuate across different task types.
 - Correlations between stress spikes and decision-making accuracy or time pressure.
 - Whether participants become more comfortable reporting biometric data over time.
 - Integration of biometric data into validated scales (stress reactivity, emotional resilience).
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9. Feedback on System Improvements

We will collect and observe participant suggestions for improving CareerQuestXR, including:

- Content scope (career options offered, explanations of recommendations).
 - User interface and VR environment enhancements.
 - Personalisation features and feedback clarity.
 - Privacy or data security concerns voiced by participants.
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10. Impact on Academic and Career Choices

Lastly, we will observe whether the use of CareerQuestXR has a measurable impact on participants' future academic and career choices. This includes:

- Whether the experience prompts them to explore new fields or adjust academic plans.
- Increased confidence in pursuing chosen career paths.
- Their likelihood of recommending VR-based career guidance to peers.